

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 transactions to guide strategic business decisions.



Used Analytical Toolkit

Leveraging industry-leading tools and platforms to transform raw data into actionable insights.



Jupyter Notebook(Python)

Robust scripting for data manipulation, statistical analysis, and predictive modeling.
Collaborative environment for data exploration, cleaning, and model development.



PostgreSQL

Secure and scalable database management for efficient data storage and retrieval.



Tableau Desktop

Interactive dashboards for compelling data visualization and reporting.

Project Overview & Dataset

Goal

Analyze spending patterns, customer segments, product preferences, and subscription behavior using 3,900 purchases across 18 columns.

Key Features

- Demographics: Age, Gender, Location, Subscription Status
- Purchase details: Item, Category, Amount, Season, Size, Color
- Behavior: Discount, Frequency, Review Rating, Shipping Type

📌 37 missing values in Review Rating column — imputed using median by product category.



Data Preparation in Python

Initial exploration used `df.info()` and `df.describe()` to check structure and summary statistics.

01

Load & Explore

Imported dataset via `pandas`; checked structure and summary statistics.

03

Feature Engineering

Created `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`.

02

Clean & Standardize

Imputed missing Review Ratings by category median; renamed columns to snake_case.

04

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.

Dataset Summary Statistics

Column	Count	Mean	Std	Min / Max
Age	3,900	44.07	15.21	18 / 70
Purchase Amount (USD)	3,900	\$59.76	\$23.69	\$20 / \$100
Review Rating	3,863	3.75	0.72	2.5 / 5.0
Previous Purchases	3,900	25.35	14.45	1 / 50

3,900

Total Transactions

18

Columns

25

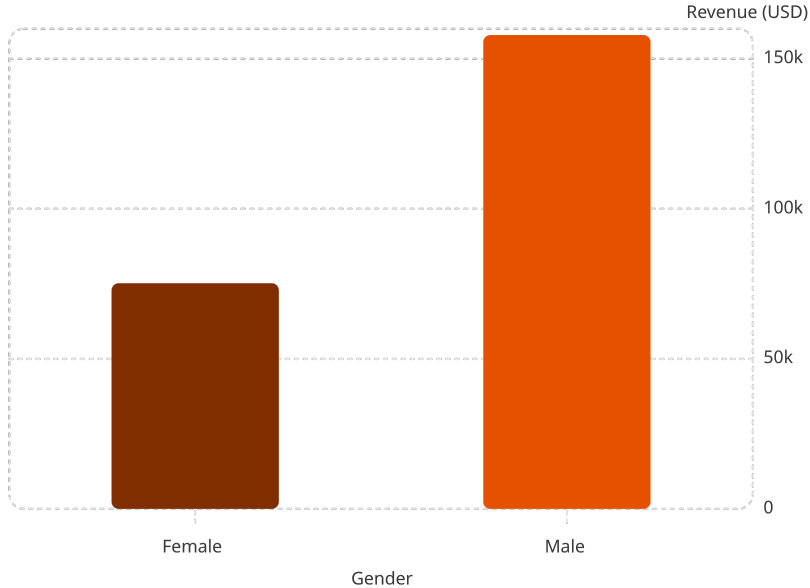
Unique Items

50

Locations

Top item: Blouse (171 purchases). Top location: Montana (96). Most common size: M (1,755). 73% male customers.

SQL Analysis — Revenue & Ratings



Revenue by Gender

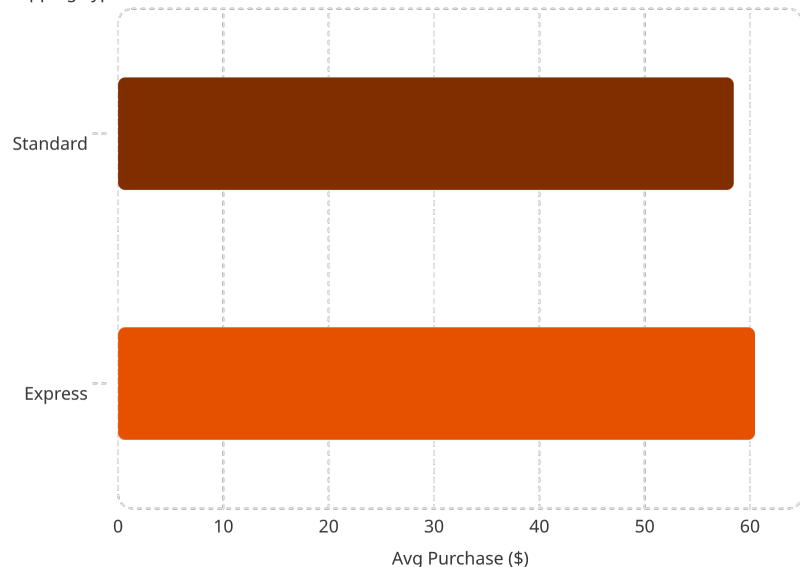
Male customers generated **\$157,890** vs. **\$75,191** for females over 2× the revenue.

Top 5 Products by Rating

1. Gloves — 3.86★
2. Sandals — 3.84★
3. Boots — 3.82★
4. Hat — 3.80★
5. Skirt — 3.78★

SQL Analysis — Shipping, Subscriptions & Discounts

Shipping Type



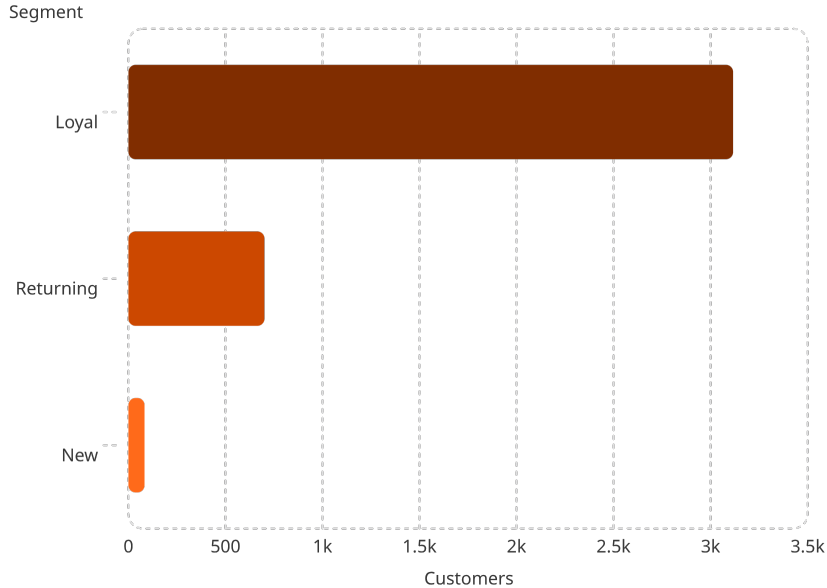
Subscribers vs. Non-Subscribers

Subscribers (27%) avg spend: **\$59.49**; Non-subscribers (73%): **\$59.87**. Total revenue: \$62,645 vs. \$170,436.

Top Discount-Dependent Products

1. Hat — 50.00% discounted purchases
2. Sneakers — 49.66%
3. Coat — 49.07%
4. Sweater — 48.17%
5. Pants — 47.37%

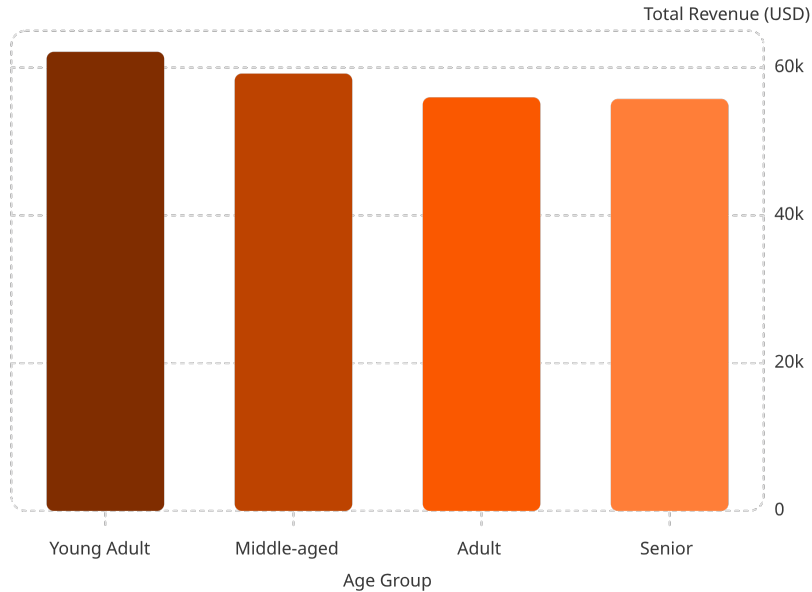
Customer Segmentation & Top Products



Top Products by Category

- **Accessories:** Jewelry (171), Sunglasses (161), Belt (161)
- **Clothing:** Blouse (171), Pants (171), Shirt (169)
- **Footwear:** Sandals (160), Shoes (150), Sneakers (145)
- **Outerwear:** Jacket (163), Coat (161)

Repeat Buyers & Revenue by Age



Repeat Buyers & Subscriptions

Of customers with **>5 purchases**: 2,518 are non-subscribers vs. 958 subscribers — a significant untapped conversion opportunity.

Revenue by Age Group

Young Adults lead with **\$62,143** in revenue, followed closely by Middle-aged (\$59,197). Revenue is fairly evenly distributed across all four age groups.

Dashboard Analysis

Key Metrics

3.9K customers · \$59.76 avg purchase · 3.75 rating

Subscription Split

27% subscribers · 73% non-subscribers

Category & Age Views

Revenue and sales by category and age group

Interactive Filters

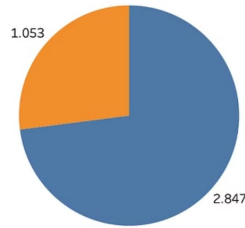
Filter by subscription, gender, category, and shipping

Tableau Dashboard

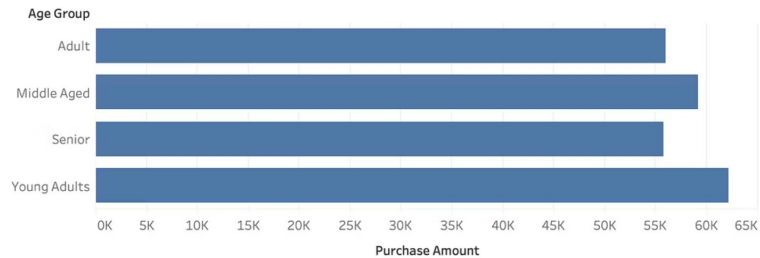
Customer Behavior Dashboard

Number of Customers

Subscription Status



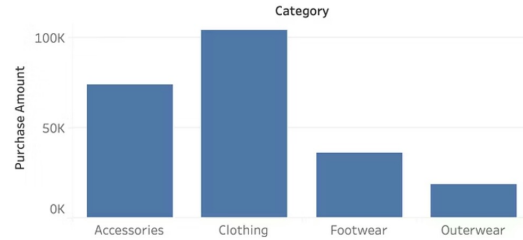
Revenue by Age Group



Average Purchase Amount

3.900

Revenue by Category

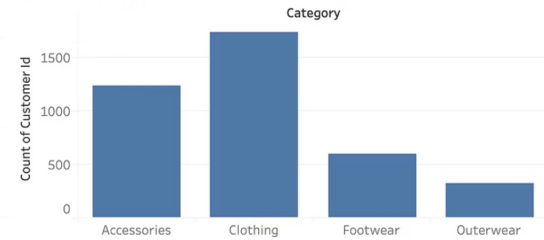


Avg Review Rating

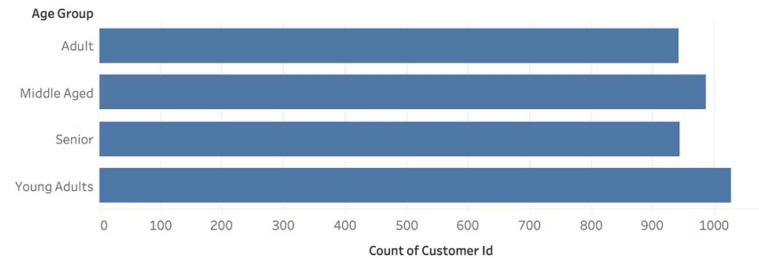
59.76

3.750

Sales by Category



Sales by Age Group



Business Recommendations

→ Boost Subscriptions

Promote exclusive benefits to convert the 2,518 repeat non-subscribers.

→ Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

→ Review Discount Policy

Balance sales boosts with margin control — 50% of Hat purchases use discounts.

→ Product Positioning

Highlight top-rated items (Gloves, Sandals, Boots) in campaigns.

→ Targeted Marketing

Focus on Young Adults and Express-shipping users for highest ROI.